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# Hydrogen transport through the tubular membranes of V-Pd alloys: Permeation, diffusion, surface processes and WGS mixture test of membrane assembly

V.N. Alimov[a], I.V. Bobylev[b], A.O. Busnyuk[a], M.E. Notkin[a,†], E.Yu. Peredistov[a], A.I. Livshits[a,*]

[a] Bonch-Bruevich Saint-Petersburg State University of Telecommunications, 22/1 prospekt Bolshevikov, St. Petersburg 193232, Russia
[b] Nizhniy Novgorod Research Institute of Radio Engineering, 5 Ul. Shaposhnikova, Nizhny Novgorod 603950, Russia

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ABSTRACT

The membranes of vanadium alloys can be a more productive and less expensive alternative to the membranes of palladium alloys for the extraction of ultrapure hydrogen. Alloying of V with Pd is very effective for the reduction of excessive hydrogen solubility inherent in pure V while the ductility of substitutional V-Pd alloys remains sufficient for making thin walled tubular membranes for practical applications. The hydrogen transport through the tubular membranes of substitutional V-$\kappa$Pd alloys with Pd coating of inner and outer sides was systematically studied with variation of alloy composition ($0 \leq \kappa \leq 13.3$ at%), temperature (300 – 500 °C) and pressure (0.001 – 1.2 MPa). The permeation flux density was almost independent of temperature for all investigated alloys. It was found that the main reason for this is the surface processes whose role dramatically increased with decreasing temperature, getting crucial at 300 °C. The record hydrogen diffusivity inherent in pure V reduced by alloying with Pd no more than 3 times even at the maximum degree of alloying (13.3%) and lowest temperature (300 °C). An assembly of 18 tubular membranes of V-5.2at%Pd with 145 μm wall thickness was designed for feeding 1 kW PEMFC and demonstrated a highly productive extraction of ultrapure hydrogen from WGS mixture.

## 1. Introduction

There are a number of important applications needing ultrapure hydrogen that could be extracted from gas mixtures with membrane technology, if highly effective and inexpensive membranes are available. First of all the generation of hydrogen from hydrocarbon fuels should be mentioned here. The global challenge is to convert the fossil fuels (including coal) into hydrogen while simultaneously concentrating and capturing $CO_2$ [1,2]. Conversion of hydrocarbon fuels into hydrogen for the electricity generation by fuel cells is another urgent direction of development [3]. Pure hydrogen production for the oil refinement is one more large-scale application where membrane technologies could be used instead of traditional PSA, if suitable membranes were available.

The Pd alloy membranes are commonly used for separating ultrapure hydrogen from gas mixtures. The commercially available self-standing membranes of Pd alloys (usually of tubular shape) have perfect selectivity but they are too expensive and insufficiently productive. The specific throughput of the supported membranes made of fine-rolled foil (≈20 μm and even thinner) is higher, but still also insufficient [3]. To enhance the throughput the composite membranes are formed by deposition of Pd- or Pd alloy films with the thickness of several microns on porous substrates. The productivity of such membranes is very high [4–6] but their selectivity and durability remain rather problematic in spite of undeniable achievements of last years.

The membranes based on group 5 metals (V, Nb, Ta) and their alloys is another option. The hydrogen transport through the lattice of group 5 metals is orders of magnitude faster than that through any other metallic lattice including the palladium one [7–14] while the transcrystalline hydrogen transfer through vanadium is fastest among the group 5 metals [10,14]. Both sides of the membrane are plated with a thin layer of palladium or palladium alloy to promote the dissociative absorption of $H_2$ molecules and associative desorption of absorbed H atoms as well as to protect the membrane from corrosion [7,8,10,15,16].

An important feature of such composite membranes is a limitation in operation temperature, which should not exceed ≈400 °C. This limitation is caused by an interdiffusion between the coating and main

* Corresponding author.
*E-mail address:* livshits@sut.ru (A.I. Livshits).
† Deceased.

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membrane material [7,8,10]. However even at the maximum admissible temperature, the hydrogen solubility in group 5 metals is so high [14,17] that the concentration of solute H at pressures of practical interest may exceed a limit permissible in terms of mechanical stability. Although the hydrogen solubility in vanadium is lowest among the group 5 metals [14,17], the equilibrium content of solute H at 400 °C exceeds atomic ratio H/M = 0.3 at feed pressure $P_{in}$ higher than 0.1 MPa and reaches almost 0.5 at $P_{in}$ = 0.3 MPa [17]. Because of so high concentration of solute H at the pressures of practical interest, pure vanadium does not seem very promising as a membrane material [18,19].

Alloying of group 5 metals allows decreasing in the hydrogen solubility, i.e. Sieverts' constant, $S$. It may seem evident that an unavoidable payment for any reduction in $S$ is a decline of the highest achievable permeation flux density, $j_m$ (due to the proportional reduction in the permeability constant). However, as demonstrated recently [19], if the membrane is made of group 5 metals or their alloys, the proper reduction of $S$ not only does not decrease $j_m$, but can lead to a *great increase* of achievable permeation flux under the typical conditions of practical operation when the outlet hydrogen pressure is about 1 atm or higher.

A number of binary and ternary alloys of vanadium with a reduced hydrogen solubility were tested in recent years and some of them demonstrated rather good properties as potential membrane materials [15,16,19–26]. The typical problem of the alloys is their insufficient ductility resulting in difficulties with the fabrication of fine rolled products (sheets or tubes). However, V–10Pd alloys were reported to have not only good permeability but also satisfactory ductility [21]. Special investigations of hydrogen solubility in substitutional V-Pd alloys [25] demonstrated that alloying of V with Pd results in very strong (maybe even record) reduction in the hydrogen solubility as well as in a number of other features favorable for using V-Pd alloys as membrane materials. Further study [19] with membranes of substitutional V-Pd alloys with different content of Pd (from 4.8 at% to 16 at%) showed that alloying of V with Pd only insignificantly reduces the hydrogen diffusivity at 400 °C and as a result a rather large permeation flux was provided by the self-standing membranes of tubular shape, which reliably operated in the pressure range of practical interest (up to 1.2 MPa) due to the admissible level of hydrogen concentration (H/M ≤ 0.2) controlled by optimum alloying.

The goal of this paper is to continue the investigation of hydrogen transport through the composite membranes based on V-Pd alloys with focusing on the temperature dependence of hydrogen permeability and diffusivity as well as on the kinetics of dissociative-associative processes at the surface of Pd coating (while PCT data for the $H_2$-(V-Pd) systems were obtained earlier [25]). Correspondingly all the parameters responsible for the hydrogen transport through the composite membranes of V-Pd alloys in the ranges of Pd concentration of 0–13.3 at% and temperatures from 300 °C to 500 °C were planned to become available.

As for the surface effects they are often ignored at consideration of hydrogen transfer through the metallic membranes. The authors believe that the paper gives a convincing illustration of a great role of surface processes specifically in the case of membranes of group 5 metals or their alloys where transcrystalline hydrogen transport is fastest. The surface effects can become crucial when temperature goes down, because the rate of dissociative absorption of $H_2$ molecules through the surface of Pd coating dramatically decreases with the temperature fall [11,29–33] while the speed of hydrogen transfer through the bulk of group-5 metals greatly increases [14] (it is especially important for the given work where temperature is the main variable parameter).

Another goal of the given work was to investigate as experimental samples the real highly productive tubular membranes of V-Pd alloys purposed to serve as modules for membrane systems of any required throughput (in contrast to a typical case when just a laboratory sample with surface area of ≈1 $cm^2$ and thickness of ≈1 mm serves as the subject of study (e.g. [15,18,20,23]). Though the alloy containing Pd is used as a material of the membrane samples, the amount of Pd is radically less than that in the Pd-based alloys (e.g. in widely used Pd-25at %Ag) applied for commercially available membranes. Besides the V-Pd membranes were found considerably more productive than Pd-alloy membranes [19]. Therefore the membranes of V-Pd alloys can be substantially more cost-effective than the self-standing membranes of Pd alloys.

Finally, if the tubular membrane samples demonstrate reliable productive operation, the membrane separator for production of ≈1 standard $m^3$ of ultrapure hydrogen per hour was planned to be assembled of such membrane units in order to extract ultrapure hydrogen from the WGS mixtures formed by the steam conversion of hydrocarbon fuels for generation of ≈1 kW power with a PEMFC battery.

## 2. Theoretical background

With taking into account the surface processes of dissociative absorption of $H_2$ molecules and associative desorption of absorbed H atoms, the correlation between the density of steady state hydrogen flux through a multilayer metallic membrane, $j$, and the inlet and outlet pressures of hydrogen ($P_{in}$ and $P_{out}$ respectively) is [7,8,19,27]

$$\sqrt{P_{in} - \frac{j}{Z_{H2}\alpha_{in}}} - \sqrt{P_{out} + \frac{j}{Z_{H2}\alpha_{out}}} - 2j\sum \frac{L_i}{S_i D_i} = 0 \quad (1)$$

where $\alpha_{in}$ is the probability of dissociative sticking of $H_2$ molecule to the inlet surface followed by hydrogen solution in the metal lattice (sticking coefficient), $\alpha_{out}$ is the same for the outlet surface, $L_i$ is the thickness of i-layer, $S_i$ and $D_i$ are the hydrogen solubility constant and diffusivity for the i-layer respectively, $Z_{H2}$ is the proportionality factor connecting hydrogen pressure and the density of incident flux (see also Notations in Appendix). Notice that $j$ in Eq. (1) is the gaseous flux expressed in $H_2$ molecules while the flux through the metallic membrane flows in the form of H atoms (i.e. it is twice larger). With the exception of the special case indicated below, Eq. (1) is valid under the condition that the concentration of hydrogen dissolved in the membrane, $C$, is in the range where Sieverts' law is valid and the diffusivities, $D_i$, do not depend on $C$.

In the particular case when the membrane is made of V-$\kappa$Pd alloy where $\kappa$ is the content of Pd in the alloy (at%) and when both sides of the membrane are covered with identical layers of Pd (i.e. the membrane is Pd-(V-$\kappa$Pd)-Pd), Eq. (1) can be specified as

$$\sqrt{P_{in} - \frac{j}{Z_{H2}\alpha}} - \sqrt{P_{out} + \frac{j}{Z_{H2}\alpha}} - 2j\left(\frac{2L_{Pd}}{S_{Pd}D_{Pd}} + \frac{L_{V-\kappa Pd}}{S_{V-\kappa Pd}D_{V-\kappa Pd}}\right) = 0 \quad (2)$$

where $S_{Pd}$, $S_{V\text{-}\kappa Pd}$, $D_{Pd}$, $D_{V\text{-}\kappa Pd}$, $L_{Pd}$ and $L_{V\text{-}\kappa Pd}$ are hydrogen solubility constants, diffusivities and thicknesses for the Pd and V-$\kappa$Pd alloy respectively. It was taken $\alpha_{in} = \alpha_{out} = \alpha$ in Eq. (2) since both sides of the membrane are identical and interact with pure hydrogen in our experiment. Notice that $\alpha$, $S_{Pd}$, $S_{V\text{-}\kappa Pd}$, $D_{Pd}$ and $D_{V\text{-}\kappa Pd}$, are functions of temperature, $T$, while $S_{V\text{-}\kappa Pd}$, and $D_{V\text{-}\kappa Pd}$ depends also on $\kappa$.

According to Eq. (2), when the condition

$$\sqrt{P_{in}} \times \alpha(T) Z_{H_2}\left[\frac{2L_{Pd}}{S_{Pd}(T)\,D_{Pd}(T)} + \frac{L_{V-\kappa Pd}}{S_{V-\kappa Pd}(\kappa, T)\,D_{V-\kappa Pd}(\kappa, T)}\right] >> 1 \quad (3)$$

is fulfilled, the permeation is limited by the bulk diffusion and $j$ can be found from Eq. (2) in an explicit form:

$$j \approx 0.5\frac{\sqrt{P_{in}} - \sqrt{P_{out}}}{\frac{2L_{Pd}}{S_{Pd}D_{Pd}} + \frac{L_{V-\kappa Pd}}{S_{V-\kappa Pd}D_{V-\kappa Pd}}}. \quad (4)$$

Vice versa, when

$$\sqrt{P_{\text{in}}} \times \alpha\,(T) Z_{\text{H}_2}\left[\frac{2L_{\text{Pd}}}{S_{\text{Pd}}(T)\,D_{\text{Pd}}(T)} + \frac{L_{V-\kappa\text{Pd}}}{S_{V-\kappa\text{Pd}}(\kappa,\,T)\,D_{V-\kappa\text{Pd}}(\kappa,\,T)}\right] << 1 \tag{5}$$

$H_2$ molecule dissociative sticking becomes the rate limiting step and $j$ is expressed as

$$j \approx \frac{1}{2}\alpha\, Z_{\text{H}_2}\,(P_{\text{in}} - P_{\text{out}})\ . \tag{6}$$

If $S_{\text{V-}\kappa\text{Pd}}(\kappa,T)$, $D_{\text{V-}\kappa\text{Pd}}(\kappa,T)$ and $\alpha\ (T)$ are known, Eq. (2) allows of calculations of any membrane system for practical use.

According to above, in general case Eqs. (1) and (2) are correct in the range of concentrations of solute H where Sieverts' law is valid and the hydrogen diffusivity does not depend on C. However this limitation is not so critical for the application of Eq. (2) because the concentration of solute H has to be anyway limited. For example, at $C$ = H/M ≈ 0.2, the ductile-to-brittle transition can be expected [18] and in addition the hydrogen dilation can result in a failure of membrane tightness. It seems to be reasonable to limit the concentration by its value of H/M ≈ 0.1 where the deviation from Sieverts' law is not very large yet, especially at elevated temperatures (> 350 °C) [25]. Besides the alloying of V with Pd results in an extension of Sieverts' law behavior to significantly higher concentration (at least up to H/M ≈ 0.3 [25]) and similar phenomenon was observed for alloying of V with Ni [24] and with Fe [28] as well. As for the diffusivity, $D$ was found to be independent of $C$ at least up to H/M ≈ 0.1 for the V-Pd alloys (for solid solutions of Pd in V) [19]. Moreover, there are grounds to believe that $D$ is independent of $C$ even at substantially higher $C$, namely, everywhere where Sieverts' law is valid. Indeed, under nondilute conditions, $D$ can be expressed as [34]

$$D = D_0 \frac{\partial \ln P^{0.5}}{\partial \ln C} \tag{7}$$

where $D_0$ is the diffusion coefficient independent of $C$, and therefore $D = D_0$, if, despite the high concentration of solute H, $C$ still continues to be proportional to $P^{0.5}$ at the equilibrium conditions [24,25,28].

Notice that the special case corresponding to condition (5) is specifically important for the given work. In this case the dissociative-associative processes at the surface are the rate limiting step of the overall permeation process and therefore the permeation flux, $j$, is almost completely determined by the dissociative sticking of $H_2$ molecules while the bulk diffusion does not play any significant role (see Eq. (6)). Therefore in the special case (5), Eq. (2) is valid even for non-dilute hydrogen solutions when Sieverts' law can be invalid and $D$ can depend on $C$. In particular, a fall of temperature can result in a drastic increase of $C$. As a result the hydrogen solution in the membrane bulk can become non-dilute and applicability of Eq. (2) can seem to be questionable. But at the same time the temperature drop can lead to a radical decrease in α resulting in the fulfilment of condition (5). If so, Eq. (2) remains to be valid in spite of any non-dilute conditions. This is just the case which will be met below.

## 3. Experimental

The membrane setup, experimental procedure, including membrane activation and other details of experiment were described elsewhere [7,8,19].

### 3.1. Sample preparation/characterization

The self-supported tubular membranes Pd-(V-κPd)-Pd were prepared on the base of seamless tubes made by OOO MEVODENA of V-κPd alloys with $0 \le \kappa \le 13.3$ at%. According to the phase diagram of V-Pd system [35] only solid solution of Pd in bcc lattice of V can form the alloy, if concentration of Pd is lower 15 at%.

The outer and inner sides of the V-κPd tubular membranes were electrolessly plated with a layer of Pd to promote dissociative/associative processes at the gas-metal interface and to protect the membrane against corrosion [7,8,19]. The thickness of the coating was from 1 to 2 μm. The plating procedure was described elsewhere [7].

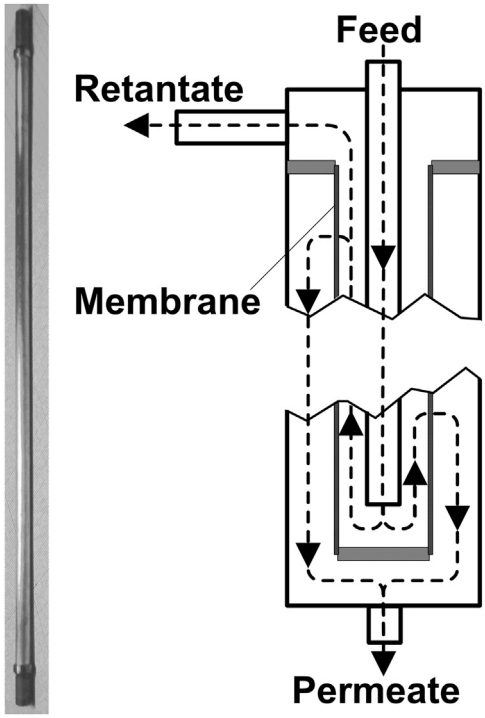



**Fig. 1.** The membrane of V-Pd alloy made of seamless tube with the inner (feed) and outer (permeate) sides plated with Pd (left) and schematic drawing of membrane in the case (right).
Both ends of the alloy tube are jointed to stainless steel by welding.

The V-κPd tubular membranes (Fig. 1) had a diameter of 0.6 cm, a length of 10–27 cm and a wall thickness of 145 μm to ≈200 μm. Both ends of the tubular membranes were arc-welded with tubes of stainless steel. One of their ends was welded to the setup while another one was dead and remained free. All the membrane samples including their welded joints were completely tight. Each sample was examined before and in the course of the permeation experiments with using the helium leak detector as well as other methods of ultrahigh vacuum technique.

The feed gas was admitted to inner side of the tubular membrane by means of thin gas-intake pipe passing inside to the dead end (Fig. 1). The membranes were placed into a thermostat with controlled temperature.

### 3.2. Measurement of the sticking coefficient

The probability of dissociative sticking of $H_2$ molecules to the surface of Pd coating followed by solution of H atoms in the metal lattice, α, was measured in the permeation experiment with the membrane of pure V. Since $S_V$ and $D_V$ are known [14,17], $\alpha(T)$ can be found from the steady state permeation flux $j(T)$ with Eq. (2). Since the transfer of H atoms through the bulk of bcc lattice of pure V is extremely fast [8,10,27], the stage of dissociative sticking markedly slows the overall permeation process down, and therefore $\alpha(T)$ can be reliably measured [7,8,19,27].

### 3.3. Measurement of the diffusion coefficient

With taking known $\alpha(T)$ and $S_{\text{V-}\kappa\text{Pd}}(T)$ one can find $D_{\text{V-}\kappa\text{Pd}}(T)$ from the experimentally measured $j(T)$ with using Eq. (2). The effects of non-dilute conditions on the accuracy of such method is discussed below.

## 4. Results and discussion

### 4.1. Permeation experiments with the membrane of pure vanadium (Pd-V-Pd)

If the subject of study is a series of binary vanadium alloys, it is natural to include the reference sample of pure vanadium in the series. However the main goal of the experiments with the Pd-V-Pd membrane was to find the probability of $H_2$ molecule dissociative sticking to Pd coating, $\alpha(T)$, with the purpose to use it for finding hydrogen diffusivity in V-κPd alloys, $D_{V\text{-}\kappa Pd}(T)$, from the experimentally measured permeation flux, $j(T)$, using Eq. (2) (Section 3.3).

The tubular membrane Pd-V-Pd with the wall thickness of $L_V$ = 189˚μm and Pd coating of $L_{Pd}$ = 2 μm was used. Two kinds of experiments were performed: the permeation flux was observed (1) at the variation of inlet ($P_{in}$) and outlet ($P_{out}$) pressures at constant temperature, $T$ = 400 °C, and (2) at the variation of temperature, $T$, at constant pressures, $P_{in}$ and $P_{out}$.

#### 4.1.1. The dependence of permeation flux on pressure

The isotherm $j_{400°C}$ was observed in two ranges of $H_2$ inlet pressure: 0.001 MPa < $P_{in}$ < 0.1 MPa and 0.1 MPa < $P_{in}$ < 1.15 MPa. In the first case, the ($N_2$+$H_2$) mixture was fed to inlet side at total pressure of 0.1 MPa, in which the $H_2$ fraction could be varied from 0% to 100%; the outlet pressure was $P_{out} \approx$ 0.001 MPa. The flux of the feed ($N_2$ + $H_2$) mixture was kept large enough to avoid the formation of noticeable gradients of hydrogen partial pressure in both the radial and longitudinal directions. In the second case, pure hydrogen at variable pressure was fed to the inlet side, and the outlet pressure was $P_{out} \approx$ 0.1 MPa. The results of these experiments are shown in Fig. 2 in the coordinates $j_{400°C}$ *vs* $(\sqrt{P_{in}} - \sqrt{P_{out}})$ that straighten the isotherm of the permeation flux in the case when the role of surface processes is insignificant (Eq. (4)).

The experimental isotherm, $j_{400°C}$, provides a clear evidence of the essential role of dissociative-associative processes in the hydrogen transfer through the Pd-V-Pd membrane. Indeed, first of all, the isotherm is no means straight line in the coordinates $j_{400°C}$ *vs* $(\sqrt{P_{in}} - \sqrt{P_{out}})$ at least in the low-pressure region (Fig. 2, ★). Secondly, *two* values of $j_{400°C}$ correspond to *one and the same* difference $\sqrt{P_{in}} - \sqrt{P_{out}}$ depending on the range of $P_{in}$. Correspondingly, two branches of $j_{400°C}$ are formed (Fig. 2) whereas, if the surface processes are insignificant (Eq. (4)), *only one* value of permeation flux can correspond to given value of difference $\sqrt{P_{in}} - \sqrt{P_{out}}$ and only single isotherm can be formed.[1]

This behavior is consistent with the theoretical concepts [7,11]: for the same value of the difference $\sqrt{P_{in}} - \sqrt{P_{out}}$, the dissociative-associative processes on the membrane surfaces reduce the permeation flux the stronger the lower the pressure (Eqs. (2)–(6)) and that leads to the splitting of the experimental isotherm into two branches as well as to the strong deviation of the lower isotherm from a straight line.

Another point is that the concentration of dissolved H becomes very high in the range of high pressures (upper branch in Fig. 2): it is only slightly lower than the equilibrium concentration which reaches H/Me = 0.52 at $P_{in}$ = 1.15 MPa [14,25]. However, the growth of $j$ with increasing $P_{in}$ does not weaken despite the high concentration (Fig. 2), and as a result a rather high density of permeation flux is reached. This would be impossible, if the hydrogen diffusivity in V noticeably drops with the increase of concentration of solute H (for example, it was reported that hydrogen diffusivity in pure Nb drops more than 3 times with the rise of H concentration [36]). Preservation of high value of $D$ under the non-dilute conditions at the hydrogen permeation through the membrane of pure V was observed also in previous experiments [8,27].

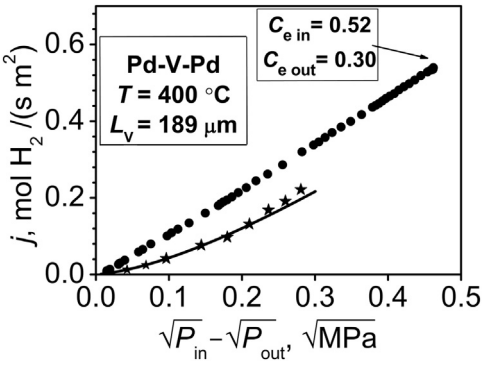



**Fig. 2.** Density of the flux permeating through the Pd-V-Pd membrane versus $\sqrt{P_{in}} - \sqrt{P_{out}}$. ★ – 0.001 MPa < $P_{in}$ < 0.1 MPa, $P_{out} \approx$ 0.001 MPa, • – 0.1 МПа < $P_{in}$ < 1.15 MPa, $P_{out} \approx$ 0.1 MPa, —— – approximation with Eq. (2) taking α = $7{\cdot}10^{-5}$. The values of the equilibrium concentrations corresponding to the maximum possible inlet ($C_{e\ in}$) and outlet ($C_{e\ out}$) pressures are indicated (in the units of atomic ratio).

#### 4.1.2. The temperature dependence of permeation flux

The permeation flux was measured at 5 values of temperatures: 300 °C, 350 °C, 400 °C, 450 °C, 500 °C. The inlet and outlet pressures of $H_2$ were constant: $P_{in}$ = 0.05 MPa and $P_{out}$ = 0.00015 MPa. To provide $P_{in}$ = 0.05 MPa the equimolecular ($H_2$ + $N_2$) mixture was fed at the inlet side at full pressure 0.1 MPa. The indicated value of $P_{out}$ was maintained by pumping. To avoid the risk of thermal degradation the membrane sample was exposed to 450 °C and 500 °C only during a limited time required for the measurement (< 20 min). Conservation of the permeability to hydrogen was examined with a standard test after any experiment at $T$ > 400 °C.

The permeation flux density, $j(T)$, found in this experiment is presented in Fig. 3. The experimental data can be compared with $j(T)$ calculated as

$$j(T) = 0.5\frac{(\sqrt{P_{in}} - \sqrt{P_{out}})\ S_V D_V}{L_V} \tag{8}$$

(solid line in Fig. 3) and as (see Eq. (4))

$$j(T) = 0.5\frac{\sqrt{P_{in}} - \sqrt{P_{out}}}{\frac{L_V}{S_V D_V} + 2\frac{L_{Pd}}{S_{Pd} D_{Pd}}} \tag{9}$$

(dash-and-dot line). It was taken for the calculation [14,17] $D_V = 3.5 \times 10^{-8}\exp(-4807(J/mol)/RT)$ $m^2/s$, $S_V = 0.24{\cdot}\exp(32646(J/mol)/RT)$ mol H/($m^3Pa^{0.5}$), $S_{Pd}D_{Pd} = 3.82 \times 10^{-7}\exp(-14000(J/mol)/RT)$ mol H/(m·s·$Pa^{0.5}$), $L_V$ = 189 μm and $L_{Pd}$ = 2.0 μm. Both equations mean that the diffusion through the bulk is the rate limiting step but Eq. (9) accounts also the diffusion through the bulk of Pd coating. Both equations are unconditionally applicable to dilute conditions though their applicability can be extended to concentrated hydrogen solutions in some cases (Sections 2 and 4.1.1). A great discrepancy between the calculations and experimental data (Fig. 3) softens with taking into account the diffusion through the Pd coating (Eq. (9)) but the difference remains to be radical (especially at lower $T$).

Two alternative explanations can be considered. First, one can suppose that the decrease in $j$ at the fall of $T$ occurs due to a great increase of concentration of solute H, $C$, (up to at least H/M ≈ 0.5, Fig. 3) resulting in approaching saturation. However, as it is demonstrated in Fig. 2 and reported earlier [8,27], $j$ by no means stops the growth proportional to $(\sqrt{P_{in}} - \sqrt{P_{out}})$ when the concentration increases becoming as high as H/M ≈ 0.5. A possible explanation can be found in Eq. (7): a deviation from Siverts' law towards a weaker

[1] Something similar to this could be observed, if the membrane was not quite tight and as a result $N_2$ would permeate along with $H_2$. However the membrane was completely tight initially and kept its tightness in the course of the permeation experiment.

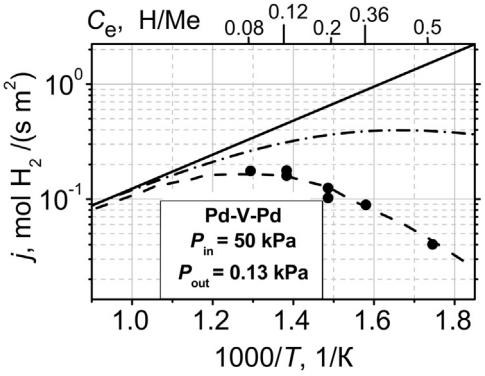



**Fig. 3.** Temperature dependence of permeation flux density through the Pd-V-Pd membrane ($L_V$ = 189 μm, $L_{Pd}$ = 2.0 μm) at $P_{in}$ = 0.05 MPa and $P_{out}$ = 0.00013 MPa. • – experiment, – – – – calculation for the similar membrane with taking into account the dissociative-associative processes at Pd coating (with Eqs. (2) and (10)), —— – calculation for the V membrane with $L_V$ = 189 μm *without* the Pd coating ($L_{Pd}$ = 0) at negligible role of surface processes (with Eq. (8)), ———– the same but *with* the Pd coating (with Eq. (9)). Equilibrium concentration of solute H at $P_{in}$ = 0.05 MPa is indicated at upper axis.

dependence of $C$ on $P$ (caused by approaching the saturation) leads to an *increase* of $D$, which can compensate the saturation effect.

We believe that the real cause of sharp decrease in $j$ with the fall of $T$ (Fig. 3) is an inhibition of the overall permeation process by the steps of dissociative absorption and associative release through the surface of Pd coating [7,8,11,27]. The double branch isotherm $j_{400^oC}$ *vs* $(\sqrt{P_{in}} - \sqrt{P_{out}})$ (Fig. 2) is a forcible argument in favor of an important role of the surface processes. Another reason is that when temperature falls, the probability of dissociative sticking of $H_2$ molecules, α, dramatically decreases [11,29–33] whereas the permeability constant, $S_V D_V$, sharply increases [14]. Therefore the role of surface processes is the greater, the lower $T$ (in accordance with Eq. (2)). Thus, if the surface processes play a significant role at 400 °C (Sec. 4.1.1, Fig. 2), their role should be even more pronounced at 350 °C and especially at 300 °C.

#### 4.1.3. The probability of $H_2$ molecule dissociative sticking to Pd coating

Thus one should conclude that the dissociative sticking of $H_2$ molecules to Pd coating significantly inhibits the permeation process and therefore α can be found with satisfactory accuracy from the experimental values of $j$ at each investigated $T$ (300, 350, 400, 450 and 500 °C). The sticking coefficients to the inlet and outlet surfaces are assumed to be equal ($\alpha_{in} = \alpha_{out} = \alpha$) since both sides of the membrane are identical (plated with Pd) and interact with pure hydrogen or nitrogen in our experiment.

First, α was found from the experimental values of $j$ *vs* $T$ (Fig. 3) with Eq. (2) – open circles in Fig. 4. Notice that according to the above

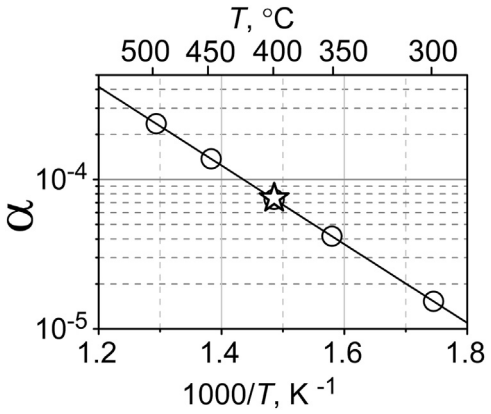



**Fig. 4.** The probability of $H_2$ molecule dissociative sticking to Pd coating.

(Section 2), Eq. (2) can be applied for finding α not only at higher $T$ (e.g. at 450 and 500 °C) where concentration of solute H, $C$, is not very high and Sieverts' law is nearly fulfilled [14,17] (the values of $C$ are indicated at the upper axis in Fig. 3). But Eq. (2) can be applied also at the lowest temperatures in spite of very high $C$ (e.g. $C$ = H/M ≈ 0.5 at 300 °C, Fig. 3). This is because the experimental values of $j$ are substantially smaller here than the values of $j$ calculated with Eq. (9) (e.g. by an order of magnitude at 300 °C, Fig. 3), and therefore the permeation flux is almost completely determined by the dissociative-associative processes at the surface of Pd coating, i.e. condition (5) is fulfilled and Eq. (2) is nearly reduced to Eq. (6), in which the bulk transport characteristics ($S$ and $D$) disappear and as a result the non-dilute state of hydrogen solution no longer plays a role.

The 400 °C isotherm of $j$ (Fig. 2) gives an additional possibility to find α(400 °C) from its low pressure branch (0.001 MPa < $P_{in}$ < 0.1 MPa). In view of low H concentration, Eq. (2) is unconditionally applicable to these data. Besides the effect of surface processes on the permeation flux is specifically strong here (Section 4.1.1) and due to this α(400 °C) can be found with a good accuracy. Technically, α(400 °C) was found by fitting the experimental data presented in Fig. 2 (lower branch) to Eq. (2) (solid line in Fig. 2). The best approximation was reached with α(400 °C) = $7{\cdot}10^{-5}$. This value is presented by the open star in Fig. 4.

As expected the experimental values of α($T$) are well approximated by a straight line in Arrehnius coordinates (Fig. 4); α($T$) can be presented as

$$\alpha(T) = 0.6 \exp\left(-\frac{50200 J/mol}{RT}\right) \quad (10)$$

Of course, the obtained α($T$) is no means a fundamental property of $H_2$-Pd system. For comparison, the kinetics of $H_2$ molecule dissociative absorption through the atomically clean surface of polycrystalline Pd is radically different: α does not depend on $T$ and close to 1 [7,8,11,27,29]. Thus α($T$) expressed by Eq. (10) relates to the particular experimental conditions and is purposed to be applied for analysis of the experimental data obtained under the same conditions (see below). On the other hand, the magnitude and temperature dependence of obtained α($T$) is typical for any non- ultrahigh vacuum conditions when the surface of Pd is inevitably covered by a film of nonmetal impurities (usually about 1 monolayer in thickness) [7,8,11,27,29–33]. The most usual contamination is carbon. The effectiveness of activation by heating in $O_2$ [29] confirms this.

If we believe that the obtained α($T$) is typical for the real operation conditions, the temperature dependence of the flux permeating through Pd-V-Pd membrane (Fig. 3) can be considered from the point of view of effect of surface kinetics. The lower is the temperature, the greater the effect. At temperatures of around 400 °C (typical operation temperature for the membranes of group 5 metals and their alloys [7,8,10,15,16,18–24]), the dissociative-associative processes inhibit the permeation through the 189 μm Pd-V-Pd membrane noticeably but not dramatically (Fig. 3). However this inhibition becomes radical at ≈300 °C. As a result when $T \leq$ 300 °C the thickness of Pd-V-Pd membrane can be increased at least by several times without any noticeable reduction in the permeation flux. Notice that the effect of surface processes on the permeation flux depends also on pressure (Eqs. (2)–(6)): the higher $P_{in}$, the smaller the role of surface processes at given α.

### 4.2. Permeation experiments with the membranes of V-Pd alloys

The permeation flux, $j$, through the membranes of Pd-(V-4.3Pd)-Pd, Pd-(V-8.9Pd)-Pd and Pd-(V-13.3Pd)-Pd are presented here depending on temperature and pressure. With the *PCT* data for the substitutional V-Pd alloys [25] as well as with the obtained above data on the kinetics of dissociative-associative processes on the surface of Pd coating (α($T$)), experimental data on $j(T, P)$ allows of finding the hydrogen diffusivity $D$

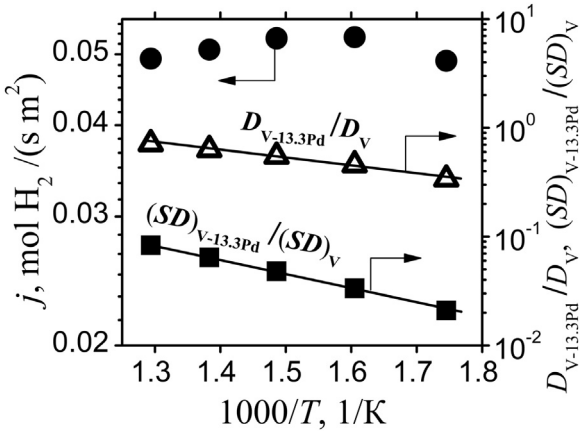



**Fig. 5.** Temperature dependence of density of the flux, $j$, permeating through the Pd-(V-13.3Pd)-Pd membrane (left axis) and of hydrogen diffusivity, $D_{V\text{-}13.3Pd}$, and permeation rate constant, $(SD)_{V\text{-}13.3Pd}$, normalized to these quantities for pure vanadium $D_V$ and $(SD)_V$ respectively (right axis).

in the V-Pd alloys depending on $T$. As a result a full set of data required for calculation of speed of hydrogen transport through the membranes Pd-(V-$\kappa$Pd)-Pd can be obtained.

Three factors complicate the determination of $D$ by the indicated way: (1) some doubts in a complete identity of surface of Pd coating for different membrane samples as well as in a complete stability of the surface state in the course of experiments, (2) deviations from Sieverts' law and (3) possible dependence of $D$ on the concentration of solute H. Factors (2) and (3) are connected with the non-dilute conditions that take place in some of our experiments.

The significance of all the three mentioned above factors is the smaller, the smaller the solubility ($S$) and permeability ($SD$) constants. Correspondingly, the experiments with the most alloyed material V-13.3Pd allow to find $D(T)$ with the highest accuracy and will be considered first.

#### 4.2.1. The alloy V-13.3Pd

The tubular membrane Pd-(V-13.3Pd)-Pd had the wall thickness $L$ = 244 μm and Pd coating thickness $L_{Pd}$ = 1.74 μm. The experimental dependence of $j$ on $T$ at $P_{in}$ = 0.58 MPa and $P_{out}$ = 0.1 MPa is presented in Fig. 5. In Fig. 6 $j$ is plotted vs $\sqrt{P_{in}} - \sqrt{P_{out}}$ at $T$ = 400 °C and $P_{out}$ = 0.1 MPa. The experimental values of $j$ at different $T$ are also presented in Table 1 along with the equilibrium concentrations of solute H ($C_{e\ in}$) corresponding to inlet pressures (taken from *PCT* data [25]) which demonstrate that a dilute solution takes place at all temperatures including the lowest one (300 °C). In Table 1 the

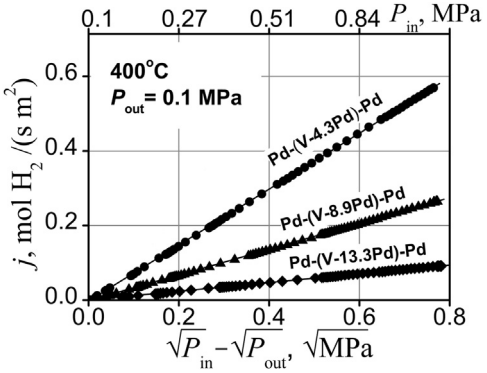



**Fig. 6.** Density of the flux permeating through the membranes Pd-(V-$\kappa$Pd)-Pd ($\kappa$ = 4.3, 8.9 and 13.3 at%) versus $\sqrt{P_{in}} - \sqrt{P_{out}}$ at $T$ = 400 °C and $P_{out}$ = 0.1 MPa. The wall thickness of tubular membranes are 203 μm, 211 μm and 244 μm for κ = 4.3, 8.9 and 13.3 at% respectively. $P_{in}$ varies from 0.1 to 1.15 MPa.

**Table 1**
Data for the membrane Pd-(V-13.3Pd)-Pd at $P_{in}$ = 0.58 MPa, $P_{out}$ = 0.1 MPa.

| $T$ °C | $J$ $\frac{mol\ H_2}{m^2 s}$ | $j_s$ $\frac{mol\ H_2}{m^2 s}$ | $S_{V\text{-}13.3Pd}$ $\frac{mol\ H}{m^3 Pa^{0.5}}$ [25] | $C_{e\ in}$ H/M | $D_{V\text{-}13.3Pd}$ $10^{-8} \times \frac{m^2}{s}$ |
|---|---|---|---|---|---|
| 500 | 0.049 | 8.2 | 4.86 | 0.029 | 1.18 |
| 450 | 0.051 | 4.3 | 5.97 | 0.036 | 1.0 |
| 400 | 0.052 | 2.4 | 7.57 | 0.045 | 0.81 |
| 350 | 0.0530 | 1.2 | 9.98 | 0.059 | 0.63 |
| 300 | 0.049 | 0.55 | 13.8 | 0.082 | 0.43 |

experimental $j(T)$ is also compared with the density of permeation flux $j_s(T)$ through a hypothetical infinitely thin membrane whose permeability is completely determined by the dissociative-associative processes at the Pd coating and correspondingly can be calculated with Eqs. (6) and (10). Since $j_s >> j$ at all temperatures (Table 1), the dissociative-associative processes have almost no effect on the observed permeation flux. As a result the condition (3) is well satisfied and Eq. (2) is nearly equivalent to Eq. (9) and therefore $j$ should be nearly proportional to $\sqrt{P_{in}} - \sqrt{P_{out}}$ at all investigated temperatures (as it was experimentally observed for 400 °C, Fig. 6).

Thus no one of mentioned above complicating factors plays a noticeable role in the case of Pd-(V-13.3Pd)-Pd membrane and $D_{V\text{-}13.3Pd}(T)$ can be found with a good accuracy from the experimental $j(T)$ either directly with Eqs. (2) and (10) or with reduced Eq. (9). The obtained values of $D_{V\text{-}13.3Pd}$ at different $T$ are presented in Table 1. It was taken for the calculation $L$ = 244 μm, $L_{Pd}$ = 1.74 μm, the values of $S_{V\text{-}13.3Pd}(T)$ [25] are given in Table 1. The hydrogen diffusivity for V-13.3 Pd alloy can be presented as

$$D_{V-13.3Pd} = 2 \times 10^{-7} \exp\left(-\frac{17900\ J/mol}{RT}\right)\ m^2/s \quad (11)$$

#### 4.2.2. The alloy V-8.9Pd

The tubular membrane Pd-(V-8.9Pd)-Pd had the wall thickness $L$ = 211 μm and Pd coating thickness $L_{Pd}$ = 2.14 μm. The experimental dependence of $j$ on $T$ at $P_{in}$ = 0.1 MPa and $P_{out}$ = 0.00145 MPa is presented in Fig. 7. Table 2 provides the data related to this experiment.

One can see (Table 2) that the equilibrium concentration at $P_{in}$ = 0.1 MPa ($C_{e\ in}$) corresponds to a dilute solution over the whole temperature range investigated and therefore Eq. (2) can be unconditionally applied as in the previous case. However, in contrast to the case of V-13.3Pd alloy, the condition $j_s >> j$ is fulfilled only at higher temperatures (≥ 450 °C) but even here far not as good as for the V-13.3Pd alloy. Thus the surface processes have noticeable effect on $j$ and therefore they give an appreciable contribution to the determination of $D_{V\text{-}8.9Pd}$ from the experimental $j(T)$ with Eqs. (2) and (10). The obtained values of $D_{V\text{-}8.9Pd}$ at different $T$ are given in Table 2.

The experimental dependence of $j$ on $\sqrt{P_{in}} - \sqrt{P_{out}}$ at $T$ = 400 °C is presented in Fig. 6. The value of $D_{V\text{-}8.9Pd}$ at 400 °C that provides the best approximation of isotherm $j_{400\ °C}$ *vs* $(\sqrt{P_{in}} - \sqrt{P_{out}})$ is also presented in Table 2 (marked with the asterisk). Notice that the isotherm $j_{400\ °C}$ *vs* $(\sqrt{P_{in}} - \sqrt{P_{out}})$ is nearly a straight line passing through the

**Table 2**
Data for the membrane Pd-(V-8.9Pd)-Pd at $P_{in}$ = 0.1 MPa, $P_{out}$ = 0.0014 MPa.

| $T$ °C | $J$ $\frac{mol\ H_2}{m^2 s}$ | $j_s$ $\frac{mol\ H_2}{m^2 s}$ | $S_{V\text{-}8.9Pd}$ $\frac{mol\ H}{m^3 Pa^{0.5}}$ [25] | $C_{e\ in}$ H/M | $D_{V\text{-}8.9Pd}$ $10^{-8} \times \frac{m^2}{s}$ |
|---|---|---|---|---|---|
| 550 | 0.072 | 2.3 | 9.13 | ≤0.015 | 1.34 |
| 500 | 0.076 | 1.7 | 11.6 | 0.026 | 1.16 |
| 450 | 0.078 | 0.90 | 15.3 | 0.033 | 1.01 |
| 400 | 0.081 | 0.50 | 20.6 | 0.044 | 0.90<br>0.80* |
| 350 | 0.076 | 0.25 | 29.4 | 0.06 | 0.83 |

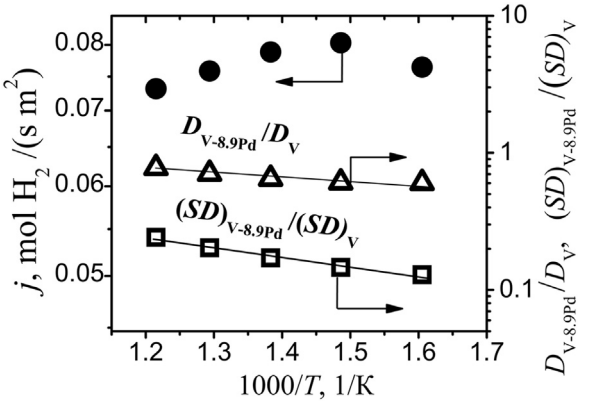



**Fig. 7.** Temperature dependence of density of the flux, $j$, permeating through the Pd-(V-8.9Pd)-Pd membrane (left axis) and of hydrogen diffusivity, $D_{V\text{-}8.9Pd}$, and permeation rate constant, $(SD)_{V\text{-}8.9Pd}$, normalized to these quantities for pure V, $D_V$ and $(SD)_V$ (right axis).

origin of coordinates (Fig. 6). This argues for an insignificant role of surface processes at 400 °C but that does not contradict to the said above about their noticeable role at this $T$ (Table 2), since this statement related to the significantly lower pressures $P_{in}$ and $P_{out}$, at which the dependence of $j$ on $T$ (Fig. 7) was observed (according to Eqs. (3)–(6) the lower is pressure the greater the role of surface processes).

The hydrogen diffusivity for V-8.9 Pd alloy can be presented as

$$D_{V-8.9Pd} = 7.9\cdot10^{-8}\exp\left(-\frac{11900\ J/mol}{RT}\right)\ m^2/s \quad (12)$$

The effects of alloying of V with 8.9 at% of Pd on the hydrogen diffusivity, $D_{V\text{-}8.9Pd}/D_V$, and on the permeation rate constant $(SD)_{V\text{-}8.9Pd}/(SD)_V$ are presented as Arrhenius graph in Fig. 7.

### 4.2.3. The alloy V-4.3Pd

The tubular membrane Pd-(V-4.3Pd)-Pd had the wall thickness $L$ = 203 μm and Pd coating thickness $L_{Pd}$ = 2.1 μm. The experimental dependence of $j$ on $(\sqrt{P_{in}} - \sqrt{P_{out}})$ at 400 °C is presented in Fig. 6. Fig. 8 shows temperature dependence of $j$ at $P_{in}$ = 0.1 MPa and $P_{out}$ = 0.0022 MPa. Table 3 provides the data similar to the data in Tables 1, 2.

The condition $j \ll j_s$ is not fulfilled at all temperatures (Table 3) and therefore the surface processes substantially reduce the permeation flux in the case of Pd-(V-4.3 Pd)-Pd membrane. Eq. (2) allows of taking into account the effect of surface processes but Eq. (2) is valid, if $S_{V\text{-}4.3Pd}$ and $D_{V\text{-}4.3Pd}$ do not depend on $C$. The latter is certainly takes place in the case of dilute solution, but the hydrogen solution is still not quite diluted at least at $T \le 400$ °C (e.g. H/M ≈ 0.15 at 350 °C, Table 3). However, it was found [25] for the nearly identical V-Pd alloy (V-4.8Pd) at the same temperatures (350 and 400 °C) that the hydrogen solubility follows Sieverts' law up to concentrations even significantly higher (up to at least H/M = 0.3) than that took place in our experiment (H/M ≤ 0.15, Table 3, Fig. 8). Thus one can believe that $C$ is everywhere proportional to $P^{0.5}$ and therefore, according to Eq. (7), $D_{V\text{-}4.3Pd}$ can be taken independent of $C$ (like $S_{V\text{-}4.3Pd}$). With taking into account the above arguments, $D_{V\text{-}4.3\ Pd}(T)$ was found from experimental $j(T)$ with Eqs. (2) and (10) and the values of $D_{V\text{-}4.3Pd}$ at different $T$ are presented in Table 3. Proceeding from these data, the hydrogen diffusivity for V-4.3Pd alloy can be expressed as

$$D_{V-4.3Pd} = 4.1\cdot10^{-8}\exp\left(-\frac{7200\ J/mol}{RT}\right)\ m^2/s. \quad (13)$$

The effects of alloying of V with 4.3 at% of Pd on the hydrogen diffusivity, $D_{V\text{-}4.3Pd}/D_V$, and on the permeation rate constant $(SD)_{V\text{-}4.3Pd}/(SD)_V$ are presented in Fig. 8.

**Table 3**
Data for the membrane Pd-(V-4.3Pd)-Pd at $P_{in}$ = 0.1 MPa, $P_{out}$ = 0.0022 MPa.

| $T$ °C | $j$ mol $H_2$/m$^2$s | $j_s$ mol $H_2$/m$^2$s | $S_{V\text{-}4.3Pd}$ mol H/m$^3$Pa$^{0.5}$ [25] | $C_{e\ in}$ H/M | $D_{V\text{-}4.3Pd}$ $10^{-8}\times$ m$^2$/s |
|---|---|---|---|---|---|
| 500 | 0.16 | 1.70 | 22.6 | < 0.1 | 1.34 |
| 450 | 0.18 | 0,90 | 30.9 | < 0.1 | 1.20 |
| 400 | 0.19 | 0.50 | 44.3 | 0.1 | 1.09 |
| 350 | 0.19 | 0.25 | 67.2 | 0.15 | 1.06 |

### 4.2.4. Generalization of experimental results and further discussion

The temperature dependences of hydrogen diffusivities, $D_{V\text{-}\kappa Pd}$, and of permeation rate constants, $(SD)_{V\text{-}\kappa Pd}$, for the investigated V-κPd alloys are gathered in Figs. 9 and 10 respectively.

First of all, it is seen (Fig. 9) that the hydrogen diffusivity does not greatly reduce in comparison with that in pure V (where $D$ is record) even at the highest level of alloying of V with Pd, compatible with the state of solid solution (≈15 at% [30]). This reduction is the greater the lower temperature but even at $T$ = 300 °C it does not exceed the factor 3 ($D_V/D_{V\text{-}13.3Pd} < 3$) and at 400 °C (typical operating temperature of membranes) $D_V/D_{V\text{-}13.3} < 2$ (Fig. 9). Thus the great reduction in the permeation rate constant (Fig. 10) occurs mainly due to an effective reduction in the hydrogen solubility (and this is the purpose of alloying).

Next consider the data obtained for $D_{V\text{-}\kappa Pd}$ (Fig. 9) in terms of their reliability. As discussed above some factors complicate the determination of $D_{V\text{-}\kappa Pd}(T)$ values from the experimentally measured permeation flux $j(T)$ and can affect their reliability. However, in the case of the membrane Pd-(V-13.3Pd)-Pd, the role of all these factors was demonstrated to be insignificant (Section 4.2.1), and therefore $D_{V\text{-}13.3Pd}(T)$

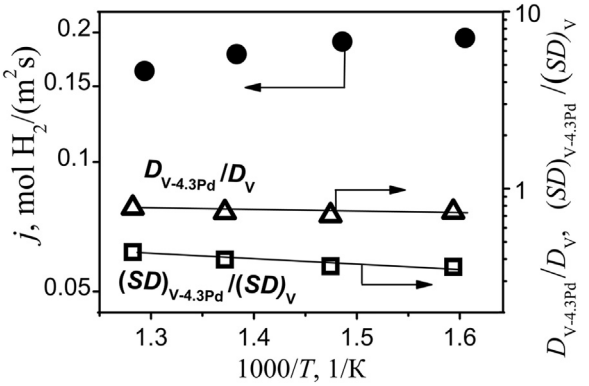



**Fig. 8.** Temperature dependence of density of the flux, $j$, permeating through the Pd-(V-4.3Pd)-Pd membrane (left axis) and of hydrogen diffusivity, $D_{V\text{-}4.3Pd}$, and permeation rate constant, $(SD)_{V\text{-}4.3Pd}$, normalized to these quantities for pure V, $D_V$ and $(SD)_V$ (right axis).

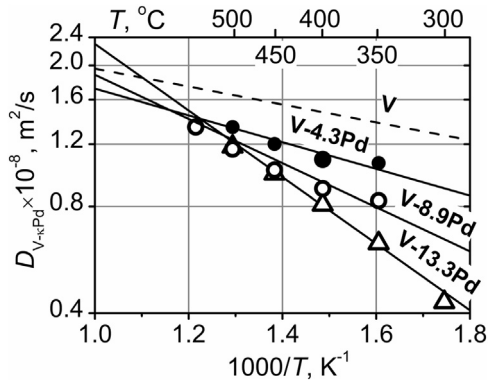



**Fig. 9.** Temperature dependence of hydrogen diffusivity in the V-κPd alloys (in solid solutions of Pd in V). The hydrogen diffusivity in pure V [14] is also presented for comparison. The solid lines correspond to Eqs. (11), (12) and (13).

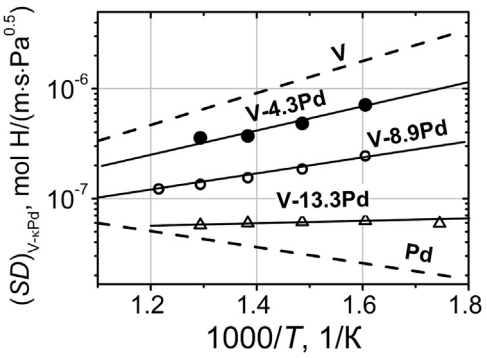



**Fig. 10.** Temperature dependence of hydrogen permeation rate constant for the V-$\kappa$Pd alloys as well as for pure V and Pd [14].

appears to be reliable. On the other hand, the diffusivities in V-13.3Pd alloy and in pure V [14] form a rather narrow corridor in the Arrehnius graph (Fig. 9). The values of diffusivity for the V-$\kappa$ Pd alloys with $0 < \kappa < 13.3$ at% are expected to be within this corridor, and the fact that they actually fall into it gives an additional argument in favor of their reliability.

Another argument in favor of the "reasonableness" of the data obtained is that they form a proper sequence. Namely, according to the experimental data (Eqs. (11)–(13), straight lines in Fig. 9), the obtained diffusivities can be expressed by the equation $D_{\mathrm{V-\kappa Pd}} = D_{0\ \mathrm{V-\kappa Pd}} \exp\left(-\frac{(E_D)_{\mathrm{V-\kappa Pd}}}{RT}\right)$, in which the higher is the level of alloying, $\kappa$, the greater the pre-exponential factor $D_{0\ V-\kappa Pd}$ and the higher the apparent activation energy $(E_D)_{\mathrm{V-\kappa Pd}}$.

In the investigated temperature range (300 – 550 °C), the higher is the level of alloying the lower is the hydrogen diffusivity (Fig. 9). However, an extrapolation of the obtained $D_{\mathrm{V-\kappa Pd}}$ $(T)$ to the range of higher $T$ indicates that this situation can become reverse, i.e. at higher $T$ the hydrogen diffusion is expected to become faster in the alloys with higher level of alloying (hypothetically it can exceed even the diffusivity in pure V). Unfortunately the membranes Pd-(V-$\kappa$Pd)-Pd can operate at $T > 400$ °C only during a short time because of limited thermal stability of Pd coating [7,18–24]. However, the membranes *superpermeable to suprathermal* hydrogen particles (e.g. to hydrogen atoms) do not have any Pd coating, and their operation temperature usually exceeds 600 °C [9,11–13]. Therefore the specifically high hydrogen diffusivity at higher temperatures could be favorable for the applications of V-Pd alloys as materials for superpermeable membranes.

The data on the apparent activation energy $(E_D)_{\mathrm{V-\kappa Pd}}$ of hydrogen diffusion (Eqs. (11)–(13)) in the combination with the data on the enthalpy of hydrogen solution, $\Delta H_{\mathrm{V-\kappa Pd}}$, [25] are presented in the form of a potential energy diagram in Fig. 11. It is seen that both the enthalpy of solution and the diffusion barrier rise with the increase of degree of

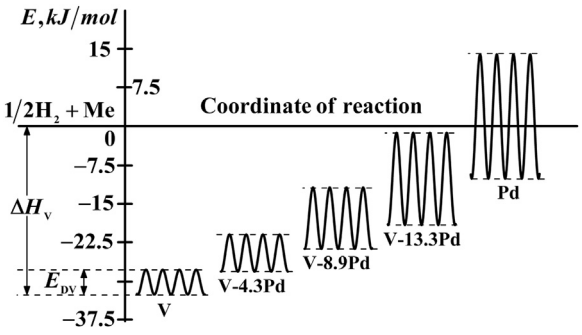



**Fig. 11.** Potential energy diagram for the several systems $H_2$-(V-$\kappa$Pd) ($\kappa$ = 0, 4.3, 8.9, 13.3, 100 at%). The energy of H atom in $H_2$ molecule is taken as the zero level. The solubility enthalpy and diffusion barrier are shown by the example of pure V ($\Delta H_V$ and $E_{DV}$ respectively).

alloying. As a result the energy level responsible for the transcrystalline hydrogen transfer, $(\Delta H + E_D)_{\mathrm{V-\kappa Pd}}$, rises with the increase of $\kappa$ but still remains substantially lower than that for pure Pd over the whole range of $\kappa$ corresponding to the state of solid solution of Pd in V.

## 5. Membrane separator for the extraction of ultrapure hydrogen from WGS mixture

The reliable and highly productive operation of tubular membranes of V-Pd alloys (Fig. 1) allowed to use them as units for a membrane system (MS) purposed for feeding a 1 kW PEM FC battery with ultrapure hydrogen extracted from the WGS mixtures being generated by catalytic steam conversion of hydrocarbon fuels. To generate 1 kW electric power at the PEM FC efficiency 42%, the MS should extract ≈13 standard liters per minute (slpm) of $H_2$ at its pressure 0.15 MPa (required to feed PEM FC battery). From the point of view of operation of both the MS and the $CH_4$ steam convertor, the optimum pressure of WGS mixture was evaluated to be ≈1.2 MPa at partial pressure of $H_2$ in the mixture 0.56 MPa. The remainder was $H_2O$, $CO_2$, CO and $CH_4$, which were approximately in proportion 1.0: 0.17: 0.23: 0.12.

To meet these conditions at operation temperature 350–370 °C the optimum content of Pd in the alloy was found to be 5.2%. In order to get the required production rate with membrane units of this alloy, the MS was assembled from 18 self-supported tubular membranes of 6 mm in diam, 27 cm in length and 145 μm in wall thickness plated with Pd. The MS was calculated to extract 80% of $H_2$ from the input WGS mixture while 84% is a maximum possible extraction ratio which could theoretically be reached under the above conditions, if the membrane area were infinite.

Such MS (Fig. 12) was assembled by OOO MEVODENA using only welding without any seals (Figs. 1 and 12). The examination with a helium leak detector as well as with other methods of ultrahigh vacuum technique showed that the MS is completely tight.

Fig. 13 (solid circles) demonstrates how the $H_2$ flux extracted by the MS changes with the increase of pressure of feed WGS mixture while its flow rate (31 slpm) and its composition remain nearly constant. The amount of $H_2$ in the feed flow was estimated to roughly correspond to the required one for generating of electric power of 1 kW scale with PEMFC (13 slpm).

It is seen that (1) the $H_2$ flux of required scale is reached (≈12 slpm), (2) this occurs just around the nominal pressure (1.2 MPa) while (3) further increase of pressure does not result in a noticeable increase of extracted $H_2$ flux. The latter means that the MS extracts the major portion of $H_2$ from the feed mixture already at nominal pressure (1.2 MPa) while the amount of $H_2$ in the feed flux is somewhat (≈10%) smaller than it is required for generating exactly 1 kW (this amount can be increased by either an increase of feed flux or partial pressure of $H_2$ in the mixture). It follows from the items (1) and (2) that throughput of the MS (i.e. total area of membrane assembly) well corresponds to the problem posed.

For an additional examination of the above conclusions, the similar experiment was carried out with the only difference that the *same* flow of the WGS mixture (31 slpm) was fed to a *single* membrane unit (but not to the 18 membrane assembly). For convenience of comparison, the flux extracted by the single membrane is presented in Fig. 13 (open circles) being multiplied by 18 (such flux is expected to be extracted with the MS, if the fed flow is increased by 18 times, i.e. up to 31 slpm × 18). As expected there is no tendency to the “saturation” behavior in this experiment since only a small portion of hydrogen can be extracted from the feed flow containing ≈12 slpm of $H_2$ by the single membrane.

The MS remained completely tight after the WGS test and therefore fully kept its capability to extract ultrapure hydrogen. It continues to operate as a part of multifuel processor in the prototype of PEMFC power source generating up to 1 kW.

Meanwhile long-term tests of the membranes of V-Pd alloys are carrying out. The duration is ≈8000 h for today and the tests are in

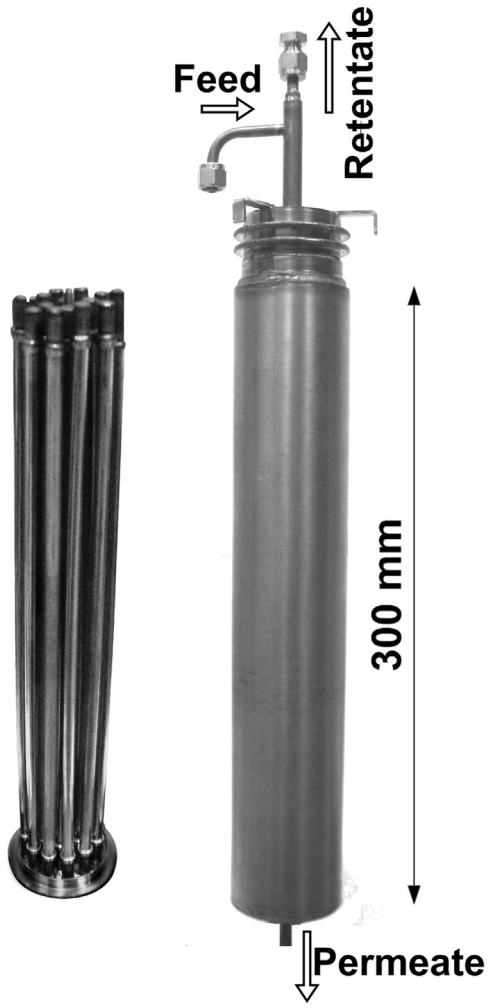



**Fig. 12.** The membrane system (MS) for feeding 1 kW PEMFC with ultrapure hydrogen. On the left: the assembly of 18 tubular Pd-(V-5.2Pd)-Pd membranes (total area is 772 $cm^2$). On the right: the MS in the case.

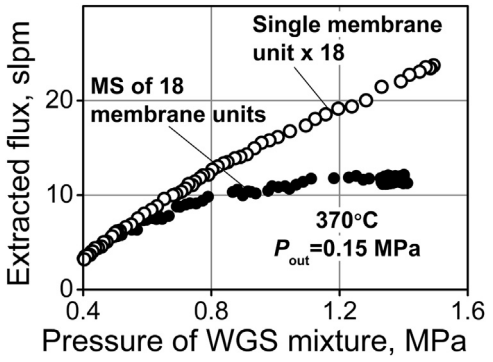



**Fig. 13.** The extracted $H_2$ flux against the pressure of WGS mixture at its nearly constant flow and composition.

progress. It should be noted in this context that operation temperature of the MS is (350–370) °C, i.e. somewhat lower than typical 400 °C. Since temperature reduction plays a crucial role for inhibiting interdiffusion between the Pd coating and bulk material [7,10,16,21,24,28], this is a key for achieving a service life acceptable for practical applications of vanadium alloy membranes (at least thousands of hours).

## 6. Conclusions

In contrast to the thick, flat and small membranes, which typically serve as laboratory samples in the studies of hydrogen transport through Group 5 metal alloys, the thin-walled tubular membranes suitable for practical applications were used as samples in this study.

The hydrogen transport through the tubular membranes of substitutional V-κPd alloys with Pd coating of inner and outer sides was systematically studied with variation of alloy composition ($0 \leq \kappa \leq$ 13.3 at%), temperature (300–500 °C) and pressure (0.001–1.2 MPa). The permeation flux density was almost independent of temperature for all investigated alloys. It was found that the main reason for this is the surface processes whose role dramatically increased with decreasing temperature, getting crucial at 300 °C. The record hydrogen diffusivity inherent in pure V reduced by alloying with Pd the greater, the lower temperature, but this reduction does not exceed factor 3 even at the maximum degree of alloying (13.3%) and lowest temperature (300 °C).

As a result of the study all the parameters responsible for the hydrogen transport through the composite membranes of V-Pd alloys in the temperature range from 300 °C to 500 °C are now available for the range of Pd content where the alloy is solid solution of Pd in V.

Since the investigated tubular membranes demonstrated reliable and highly productive operation, an assembly of 18 tubular membranes of V-5.2 at%Pd alloy with 145 μm wall thickness was designed for feeding 1 kW PEMFC with ultrapure hydrogen extracted from WGS mixtures. This membrane system was successfully tested and continued to operate as a part of multifuel processor of 1 kW PEMFC power source.

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## Appendix. Notations

$\kappa$ = content of alloying element (Pd) in the alloy (V- $\kappa$Pd),
$P$ = hydrogen pressure, $P_{in}$ and $P_{out}$ = the same at the inlet and outlet sides of membrane respectively,
$C$ = concentration of dissolved hydrogen, $C_e$ = the same for equilibrium, $C_{e\ in}$ *and* $C_{e\ out}$ = the same for equilibrium at pressure $P_{in}$ and $P_{out}$ respectively,
$j$ = permeation flux density,
$Z_{H2}$= gas kinetic theory coefficient ($Z_{H2} = \frac{3.2\ mol\ H_2}{\sqrt{T}\ m^2 s\ Pa}$),
$\alpha$ = the probability of dissociative absorption of $H_2$ molecule through the surface of Pd coating per single impact on the surface (sticking coefficient), $\alpha_{\text{in}}$, $\alpha_{\text{out}}$ = the same for the inlet and outlet sides of the membrane respectively,
$j_s$ = the density of the hydrogen flux through an infinitely thin symmetric ($\alpha_{\text{in}} = \alpha_{\text{out}} = \alpha$), membrane, $j_s = 0.5 Z_{H2} \alpha\ (P_{in} - P_{out})$,
$L_i$, $L_V$, $L_{Pd}$, $L_{V\text{-}\kappa Pd}$ = thickness of the layer consisting of the substance $i$, of V, Pd and V-κPd alloy respectively (the layers compose the membrane),
$S_i$, $S_V$, $S_{Pd}$, $S_{V\text{-}\kappa Pd}$ = the hydrogen solubility constant (Sieverts' constant) for the substance $i$, for V, Pd and V-κPd alloy respectively,
$D_i$, $D_V$, $D_{Pd}$, $D_{V\text{-}\kappa Pd}$ = the hydrogen diffusivity for the substance $i$, for V, Pd and V-κPd alloy respectively,
$(E_D)_{V-\kappa Pd}$ = the apparent activation energy of hydrogen diffusion in V-κPd alloy,
$\Delta H_{V\text{-}\kappa Pd}$ = the enthalpy of hydrogen solution in V-κPd alloy.

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